

# SWARIT TIWARI

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## SUMMARY

M.Sc. Mechanical Engineering & Management graduate with experience in CAD/CAE design, manufacturing improvement, data analysis and business-oriented engineering projects. Skilled in mechanical design, simulation, Python automation and analytical tools, with project work across product development, maritime fuels and NLP-based customer insights.

## SKILLS

**Languages:** German (B2), English (Fluent), Hindi (Native)

**Engineering Software:** CATIA 3DEXPERIENCE, Siemens NX, SolidWorks, Fusion 360, AutoCAD, ANSYS, Inventor

**Business Tools:** Power BI, Tableau, Jira, SAP ERP, Advanced Excel, MS Office

**Programming:** Python (Pandas, NumPy, transformers), MATLAB, Simulink, SQL.

## EDUCATION

10/2022 – 03/2026 **M.Sc. in Mechanical Engineering and Management**  
*Technische Universität Hamburg, Germany*

07/2017 – 07/2021 **B.Eng. in Mechanical Engineering**  
*Gujarat Technological University, India*

## PROFESSIONAL EXPERIENCE

07/2021 – 08/2022 **Jr. Design Engineer**  
*GERENT, India*

- Developed and validated mechanical components using Fusion 360, SolidWorks and ANSYS, supporting design, simulation and production readiness.
- Analyzed manufacturing processes and implemented improvements that reduced production line waste by 35%.
- Automated technical certification documentation with Python/VBA, reducing documentation effort by 75%.

06/2020 – 07/2020 **Intern**  
*GURUKRUPA KRAFTS PVT. LTD., India*

- Operated and maintained production machinery, including lathe, CNC, VMC, gear cutters, and hydraulic presses, gaining in-depth manufacturing knowledge.

## PROJECTS

**Sentiment Analysis and Topic Modelling of Restaurant Reviews | Master Thesis | Python, BERTopic, BERT** 🔗  
*Filtered 1M+ Google reviews to 212k+ restaurant reviews and extracted customer-experience insights using NLP.*

**Alternative Fuels in Maritime Industry | Fraunhofer CML Project | Power BI, Jira, Excel** 🔗  
*Developed a weighted scoring matrix to compare alternative maritime fuels across commercial and operational criteria.*

**Single-Print FDM Race Car | Fusion 360, SolidWorks, FDM, DfAM** 🔗  
*Designed a movable single-print race car with functional wheels, printability and no-assembly design.*

**Topology-Optimized 3D Printed Bridge | SolidWorks, Simulation, FFF/MEX, Testing** 🔗  
*Developed and tested a lightweight bridge, 172.6 g mass and 500+ N load capacity.*

**Contactless Sanitization Station | CAD Concept, Product Design, Autodesk Hackathon** 🔗  
*Developed a CAD concept for an automatic sanitization station, awarded 2nd place at Autodesk Hackathon.*

**Regenerative Braking System for Bicycles | CAD, ANSYS, Mechanical Design** 🔗  
*Designed and validated a modular front-wheel regenerative braking concept for bicycles.*